Idurus Mohamed
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Assignment 2

1. Prompt log

Part 1: Start simple

Prompt: Are these reviews positive or negative?

Response:The dataset contains **110 reviews** evenly distributed across 5 products (22 reviews each). Rather than leaning strictly one way, **the reviews are mixed, with negative reviews slightly edging out positive ones.**

1. Overall Sentiment Breakdown

Sentiment Category	Count	Percentage	Description / Characteristics
Negative	**39**	35.5%	Severe defects, durability failures, poor customer service, and heavy sarcasm.
Positive	**35**	31.8%	Strong endorsements, repeat purchases, and satisfaction with value.
Mixed	**22**	20.0%	Mention both clear pros and cons (e.g., great hardware, poor companion app).
Neutral / Questions	**14**	12.7%	Flat observations ("it's a kettle"), delivery notes, or customer questions.

2. Breakdown by Product

Product	Positive	Negative	Mixed	Neutral / Question	General Sentiment

Cozy Knit Throw Blanket	**8**	6	5	3	**Slightly Positive** (Soft and cozy, but shedding/shrinkage complaints)
Lumen Smart Bulbs (4-Pack)	**8**	**9**	3	2	**Polarized / Leaning Negative** (Good light quality, but unreliable app/Wi-Fi)
Trailmark Hiking Backpack	**7**	**8**	4	3	**Balanced** (Praised for ruggedness, criticized for strap comfort/zippers)
Hearth Electric Kettle	**7**	**8**	4	3	**Balanced / Safety Concerns** (Fast boil, but complaints of hot handle/leaks)
Breeze Wireless Earbuds	5	**8**	6	3	**Leaning Negative** (Decent audio value, but microphone and fit issues)

3. Key Observations & Nuances

- **Frequent Sarcasm / Irony:** Several negative reviews use positive surface words that traditional keyword-based sentiment tools frequently misclassify as positive:
    - *R022: "Love that the 'waterproof' pack turned my sleeping bag into a sponge. Ten stars."* (Negative)
    - *R030: "Fantastic. Just fantastic. Mine lasted a full week before the switch broke."* (Negative)
    - *R080: "Oh sure, I just love when the buckle snaps halfway up a mountain."* (Negative)
    - *R085: "Oh good, a kettle that makes my water taste like plastic. Exactly what I wanted."* (Negative)

- **Hardware vs. Software Duality:** In the Lumen Smart Bulbs, users frequently like the physical hardware (brightness/colors) but rate the product poorly due to connectivity drops and companion app bugs (*R008, R009, R031*).
- **Non-Product Reviews:** A handful of reviews focus on delivery carriers, wrong shipments, or questions rather than evaluating product quality (e.g., *R019: "Seller sent the wrong size," R087: "What are the dimensions?"*).

Part 2: Improve the Prompt

Technique: Define categories and Few-shot prompting:

Prompt - Are these reviews positive or negative? The categories you're going to be using are positive, negative, comments, and neutral. Positive comments are for the ones that are positive. Negative comments are for the ones that are negative. - The comments category is for anything that is not necessarily a positive or negative review. It's just a question or advice that doesn't fall under positive or negative. - For neutral, this is when it comes to neither being positive nor negative. That's when you would use neutral if you don't really know whether the comment is one of those two main categories of positive or negative. - If you see anything sarcastic, that will fall under negative. Anything sounding negative, thus using sarcasm, will fall under the negative category.So the following is an example that you can copy from, okay? So if a review said this product was so amazing that it didn't last for an hour, so I would label that as Negative, because it's using sarcasm. And then another example is, what time should I expect to get my product? That will be labeled as, in this case, that I will label it as comment, because this is a question and the user is expecting an answer, so it's neither positive or negative. I have no idea what this is about! Wait for me to take a picture of this! What is that? It's a flower! And a final example is if a user stated something along the lines of, I love this product so far, but there are some features that I'm not too sure of if it can help me. So I think I'm gonna have to keep using it to find out if it's gonna help me. So I would label that as. And as a neutral review, because they're praising it and at the same time they're saying they're not too sure because, you know, they haven't used it, and so that wouldn't be necessarily positive, and it also wouldn't be negative because it's just neutral.
Response:

Review ID	**Product**	**Review Text**	**Classification**
R001	Trailmark Hiking Backpack	Smells like chemicals out of the box. Took a week to air out.	**Negative**

R002	Breeze Wireless Earbuds	Noise isolation is decent, but it's not true noise cancelling like the description implies. Still good for the gym.	**Neutral**
R003	Breeze Wireless Earbuds	Left earbud stopped charging after two weeks. Contacted support and never heard back.	**Negative**
R004	Trailmark Hiking Backpack	Straps dig into my shoulders after an hour. Returning.	**Negative**
R005	Cozy Knit Throw Blanket	Not warm enough for winter. Fine as a decorative throw.	**Neutral**
R006	Breeze Wireless Earbuds	Perfect for my commute. Clear calls, solid battery.	**Positive**
R007	Trailmark Hiking Backpack	decent bag for the money, not for serious backpacking tho	**Neutral**
R008	Lumen Smart Bulbs (4-Pack)	Bright and colorful, but they disconnect from Wi-Fi every few days.	**Neutral**
R009	Lumen Smart Bulbs (4-Pack)	Great bulbs, terrible app. If you can get past setup you'll love them.	**Neutral**

R010	Hearth Electric Kettle	Second one I've bought, gave the first to my mom. Still going strong after a year.	**Positive**
R011	Cozy Knit Throw Blanket	Arrived with a stain on it.	**Negative**
R012	Lumen Smart Bulbs (4-Pack)	Do these work with Google Home or only Alexa?	**Comment**
R013	Hearth Electric Kettle	Stopped turning on after 6 weeks. Customer service sent a replacement, which works so far.	**Neutral**
R014	Hearth Electric Kettle	Mine arrived dented and customer service shipped a new one the next day. The kettle itself is fine.	**Neutral**
R015	Trailmark Hiking Backpack	Way too small for anything over a day trip despite what the listing says.	**Negative**
R016	Trailmark Hiking Backpack	Ordered for my husband. He says it's "fine."	**Neutral**
R017	Trailmark Hiking Backpack	Absolutely bombproof. Dragged it through brush, rocks, and rivers with zero damage.	**Positive**

R018	Lumen Smart Bulbs (4-Pack)	Colors look washed out compared to other brands I've used.	**Negative**
R019	Trailmark Hiking Backpack	Seller sent the wrong size.	**Negative**
R020	Trailmark Hiking Backpack	Color is different from the photos. More brown than green. Still a good bag.	**Neutral**
R021	Trailmark Hiking Backpack	Great value. Feels much more expensive than it was.	**Positive**
R022	Trailmark Hiking Backpack	Love that the "waterproof" pack turned my sleeping bag into a sponge. Ten stars.	**Negative** *(Sarcasm)*
R023	Lumen Smart Bulbs (4-Pack)	Sunrise alarm feature has made my mornings so much easier.	**Positive**

Technique: Output format and Constraints
Prompt: Are these reviews positive or negative? Can you format it to only include the product name, the review text, and the classification? If a review is unclear, can you just mark it as unclear.

Result:

Product Name	Review Text	Classification
Trailmark Hiking Backpack	Smells like chemicals out of the box. Took a week to air out.	Negative
Breeze Wireless Earbuds	Noise isolation is decent, but it's not true noise cancelling like the description implies. Still good for the gym.	Positive
Breeze Wireless Earbuds	Left earbud stopped charging after two weeks. Contacted support and never heard back.	Negative
Trailmark Hiking Backpack	Straps dig into my shoulders after an hour. Returning.	Negative
Cozy Knit Throw Blanket	Not warm enough for winter. Fine as a decorative throw.	Unclear
Breeze Wireless Earbuds	Perfect for my commute. Clear calls, solid battery.	Positive
Trailmark Hiking Backpack	decent bag for the money, not for serious backpacking tho	Unclear
Lumen Smart Bulbs (4-Pack)	Bright and colorful, but they disconnect from Wi-Fi every few days.	Negative

Lumen Smart Bulbs (4-Pack)	Great bulbs, terrible app. If you can get past setup you'll love them.	Positive
Hearth Electric Kettle	Second one I've bought, gave the first to my mom. Still going strong after a year.	Positive
Cozy Knit Throw Blanket	Arrived with a stain on it.	Negative
Lumen Smart Bulbs (4-Pack)	Do these work with Google Home or only Alexa?	Unclear
Hearth Electric Kettle	Stopped turning on after 6 weeks. Customer service sent a replacement, which works so far.	Unclear

Technique: Role prompting
Prompt: Take the role of client Relationship Manager. Are these reviews positive or negative?

Response:
The review dataset consists of **110 reviews across 5 product lines** (22 reviews each). Overall sentiment is **mixed and divided almost evenly**, with a slight positive skew:

- **Positive / Favorable:** ~45% (49 reviews)
- **Negative / Critical:** ~41% (45 reviews, notably including multiple highly sarcastic 1-star-style remarks)
- **Neutral / Pre-sale Inquiries:** ~14% (16 reviews)

Portfolio Sentiment Breakdown

Product	**Dominant Sentiment**	**Positive**	**Negative / Mixed Neg**	**Neutral / Inquiries**	**Primary Drivers &**

					Key Pain Points
Trailmark Hiking Backpack	**Leaning Positive**	11	8	3	**Praise:** Durability, pocket utility, value.

Concerns: Chemical odor upon unboxing, strap comfort, zipper/buckle failure. | | **Cozy Knit Throw Blanket** | **Leaning Positive** | 10 | 8 | 4 | **Praise:** Aesthetics, gift appeal, softness.

Concerns: Fuzz shedding, shrinking in wash, snags, warehouse odor. | | **Hearth Electric Kettle** | **Balanced / Split** | 11 | 9 | 2 | **Praise:** Rapid boiling, sleek gooseneck spout.

Concerns: Critical safety hazards (burning hot handle, bottom leaks near electrical outlets), switch failures. | | **Lumen Smart Bulbs (4-Pack)| Balanced / Split** | 9 | 10 | 3 | **Praise:** Color vibrance, Alexa integration, sunrise alarm.

Concerns: Wi-Fi disconnects, mobile app crashes, aggressive account/location permissions. | | **Breeze Wireless Earbuds** | **Leaning Negative** | 8 | 10 | 4 | **Praise:** Sound/bass value for price, battery life.

Concerns: One earbud failing to charge, poor microphone quality, unresponsive support (ghosting). |

Relationship Manager's Executive Assessment

- **Urgent Risk (Safety & Liability):** The **Hearth Electric Kettle** has multiple reports of severe functional failures, notably **handles burning users' hands** (R049) and **water leaking from the base near electrical outlets** (R048). These require immediate quality assurance escalation to prevent product liability or recall scenarios.
- **Customer Support Bottlenecks:** While some customers commended swift replacements (R013, R014), others experienced severe lapses, such as **unanswered tickets for hardware failure** (R003). Standardizing SLA response times is crucial for customer retention.
- **Listing Expectation Gaps:** A notable fraction of negative reviews stems from misalignment between product marketing copy and actual performance—specifically

"waterproof" claims on the backpack that resulted in soaked gear (R022), "noise isolation" branded as active noise cancellation (R002), and throw blanket washing care specifications (R071). Updating listing copy will curb preventable returns.

Part 3: Check the model's work

2. Spot-Check Table

Review ID	Review Text	My Label	Model's Label	Agreed ?
R013	Stopped turning on after 6 weeks. Customer service sent a replacement, which works so far.	Negative	Neutral	No
R014	Mine arrived dented and customer service shipped a new one the next day. The kettle itself is fine.	Negative	Neutral	No
R015	Way too small for anything over a day trip despite what the listing says.	Negative	Negative	Yes
R016	Ordered for my husband. He says it's "fine."	Positive	Neutral	Yes
R017	Absolutely bombproof. Dragged it through brush, rocks, and rivers with zero damage.	Positive	Positive	Yes
R018	Colors look washed out compared to other brands I've used.	Negative	Negative	Yes
R019	Seller sent the wrong size.	Negative	Negative	Yes
R020	Color is different from the photos. More brown than green. Still a good bag.	Neutral	Neutral	Yes
R021	Great value. Feels much more expensive than it was.	Positive	Positive	Yes
R023	Sunrise alarm feature has made my mornings so much easier.	Positive	Positive	Yes

3. Analysis

When I started simple in Part 1 by just asking if the reviews were positive or negative, the model gave me mostly summarizations and a breakdown by product instead of labeling each review

one by one. For reviews that were not clearly positive or negative, it created its own categories of mixed and neutral/questions.

In Part 2, I used different techniques to improve the prompt and saw how each one changed the output. First, I used defined categories and few-shot prompting by telling the model to use positive, negative, comments, and neutral, and giving it examples to copy from. With this technique, the model stopped summarizing and labeled every single one of the 110 reviews individually, though it did add "(Sarcasm)" in parentheses next to a negative label when I didn't ask it to.

Next, I used output format and constraints by telling it to only include the product name, review text, and classification without the product ID, and to mark unclear reviews as "Unclear," which it followed.

Finally, I used role prompting by telling it to act as a Client Relationship Manager. This completely changed the tone into business language, giving me stats and highlighting urgent safety risks, customer support bottlenecks, and listing expectation gaps instead of individual labels.

When I checked the model's work against my 10 hand-labeled reviews, we agreed on everything except two tricky ones. At first, I wondered if the review that said "Ordered for my husband. He says it's 'fine'" was positive, because there was no negative sentiment, while the AI put neutral, and after looking at why "fine" was in quotation marks, I agreed that neutral was right.

However, for the kettle review where it stopped turning on after six weeks and the one where the kettle arrived dented, I put negative and the AI put neutral. I think I am right on both of those because the customer is already unhappy and having a bad experience, even if customer service sent a replacement. To fix this, I updated my prompt to tell the model that those two reviews were clearly negative and needed to be fixed.

Overall, the model did label every single review when I used defined categories and few-shot prompting, and that technique made the biggest difference for me. Because that prompt was in-depth and gave examples of what I expected, it gave me all 110 reviews labeled individually, showing me how important these techniques are for getting exactly what you want from AI instead of something generic.